ELEC 221 Lecture 16
The DT Fourier transform

Tuesday 3 November 2026

Announcements

- No quiz today!

Where are we going?

Today

Learning outcomes:

- Define and compute the DT Fourier transform for aperiodic signals
- Use the convolution property of the DTFT to analyze the behaviour of LTI systems
- Construct and analyze DT systems based on difference equations

Recap: DT Fourier series

We can express a periodic DT signal (period N) as a discrete Fourier series.

DT synthesis equation:

DT analysis equation:

The DT Fourier transform

The discrete-time Fourier transform (DTFT) is the generalization of the discrete Fourier series to **aperiodic** signals.

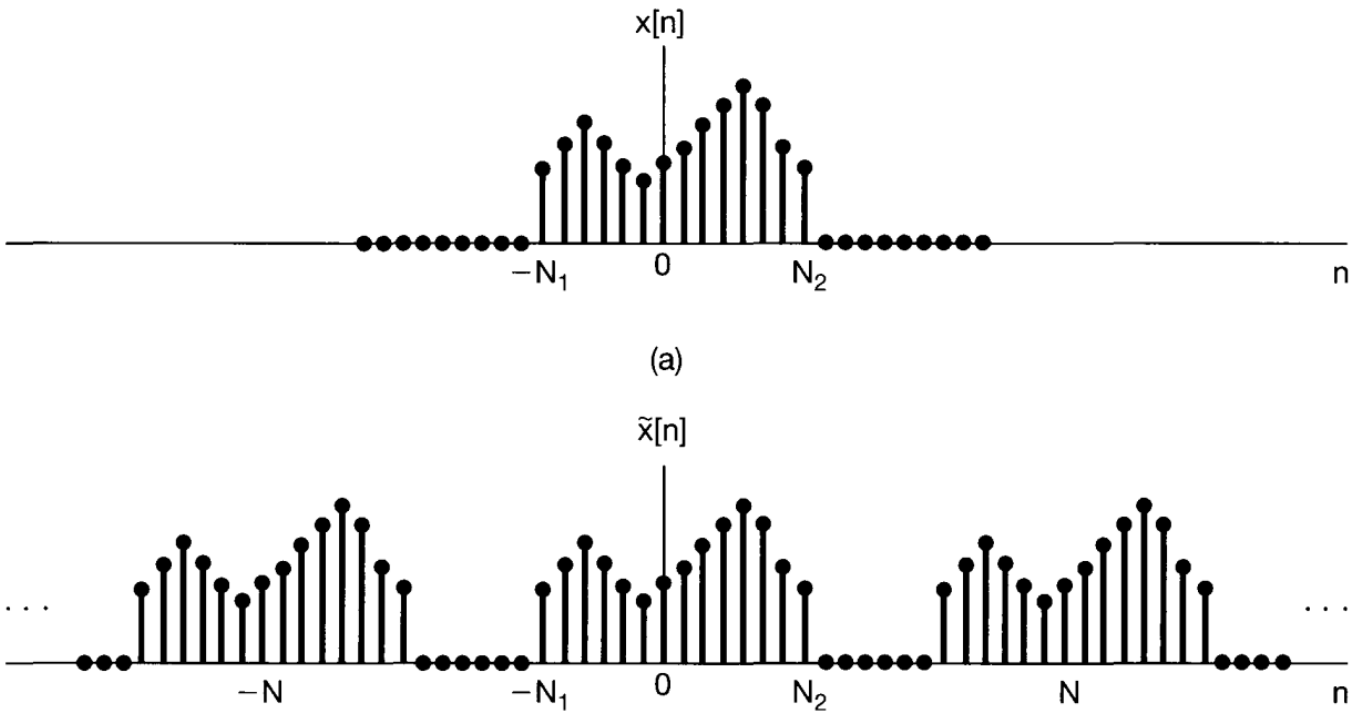


Image credit: Oppenheim chapter 5.1

The DT Fourier transform

Suppose $\tilde{x}[n]$ is a periodic extension of $x[n]$.

Set the bounds to consider where our signal actually is:

What happens if we increase the period?

The DT Fourier transform

$$c_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} \tilde{x}[n] e^{-jk\frac{2\pi n}{N}}$$

If $N \to \infty$, for any finite n, $\tilde{x}[n]$ looks just like $x[n]$:

Since $x[n] = 0$ outside this range, we can extend the bounds:

The DT Fourier transform

We have

Define

The DT Fourier transform

Substituting

$$c_k = \frac{1}{N} X(e^{jk\omega})$$

into the synthesis equation for $\tilde{x}[n]$ yields

What happens as $N \to \infty$?

Over what range should we integrate ω?

The DT Fourier transform

DT Fourier transform pair:

Inverse DTFT (synthesis equation)

DTFT (analysis equation)

Example: DTFT of a square pulse

Compute the DTFT of the DT signal

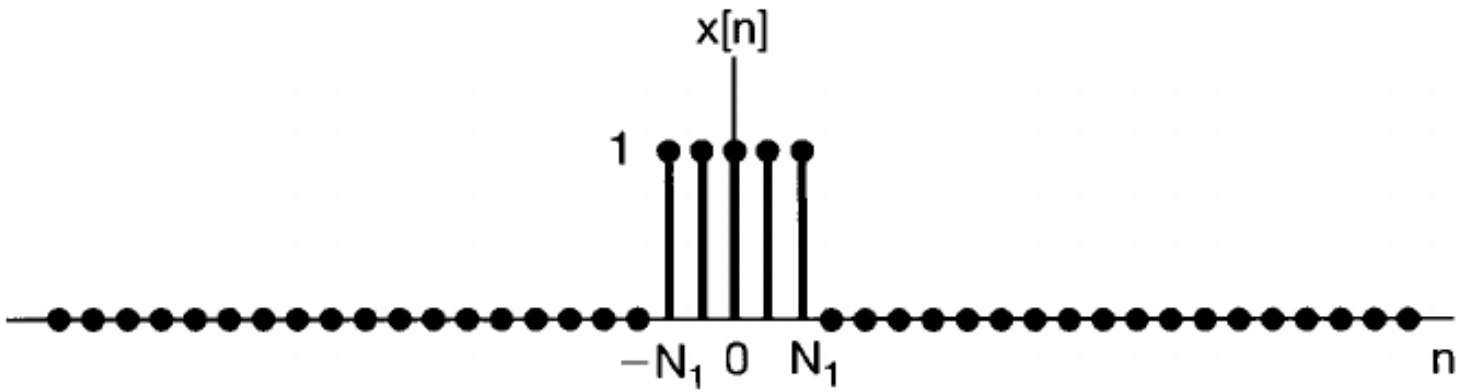


Image credit: Oppenheim chapter 5.1

Recall: FT of a CT square pulse

$$x(t) = \begin{cases} 1 & |t| < T_1, \\ 0 & |t| > T_1 \end{cases}$$

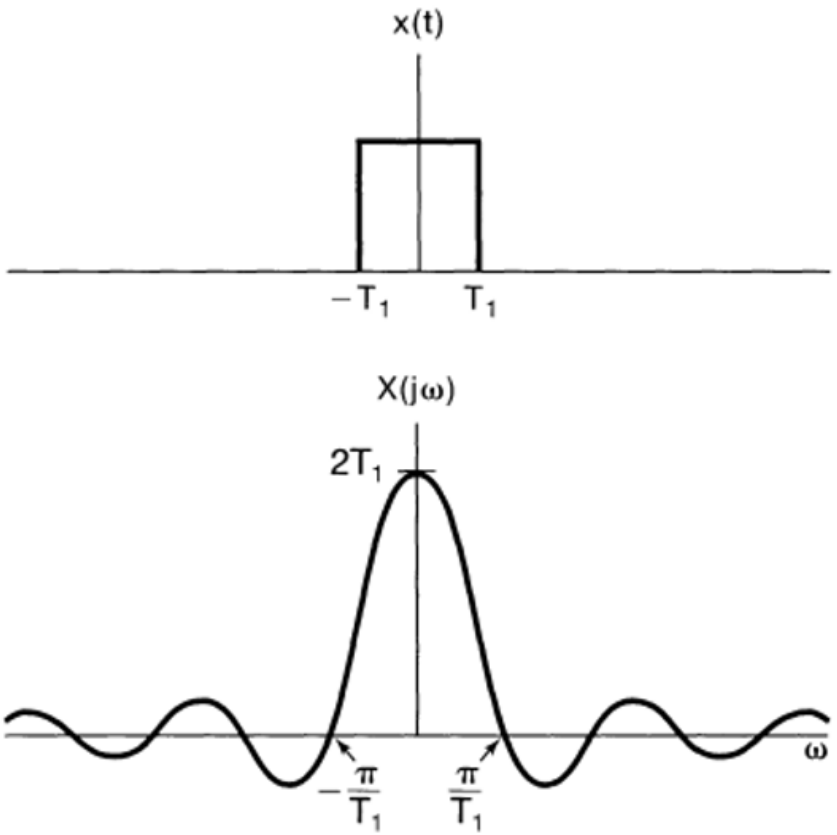


Image credit: Oppenheim chapter 4.1

Example: DTFT of a square pulse

$$x[n] = \begin{cases} 1 & |n| \leq N_1 \\ 0 & |n| > N_1 \end{cases}$$

Compute the DTFT:

How do we evaluate this sum?

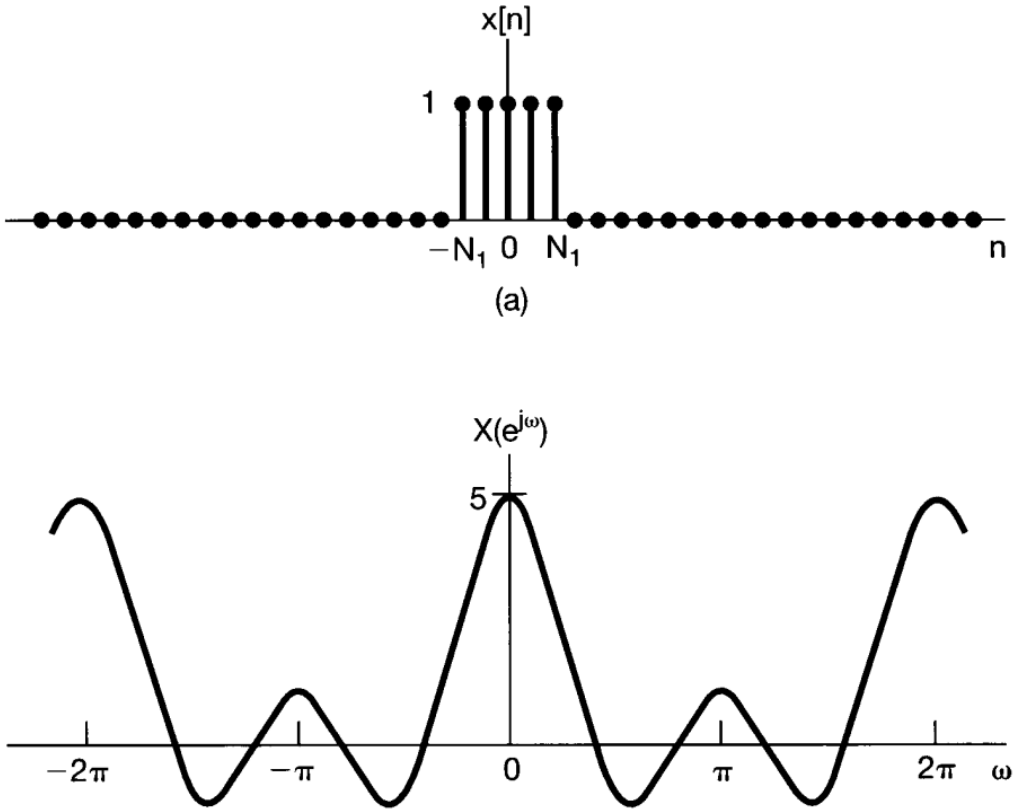
Example: DTFT of a square pulse

Change summation variable to $m = n + N_1$

Use our handy identity:

Do some reshuffling...

Example: DTFT of a square pulse



Note this function is **continuous** and **periodic**!

Image credit: Oppenheim chapter 5.1

Convergence criteria

In CT had Dirichlet criteria for both Fourier series and inverse Fourier transform. No conditions for DT Fourier series:

$$x[n] = \sum_{k=\langle N \rangle} c_k e^{jk\omega n} \qquad c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega n}$$

What about the DT Fourier transform?

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \qquad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Convolution

DTFT defines the relationship between impulse response and frequency response:

Convolution works the same way as in CT:

Example: filters

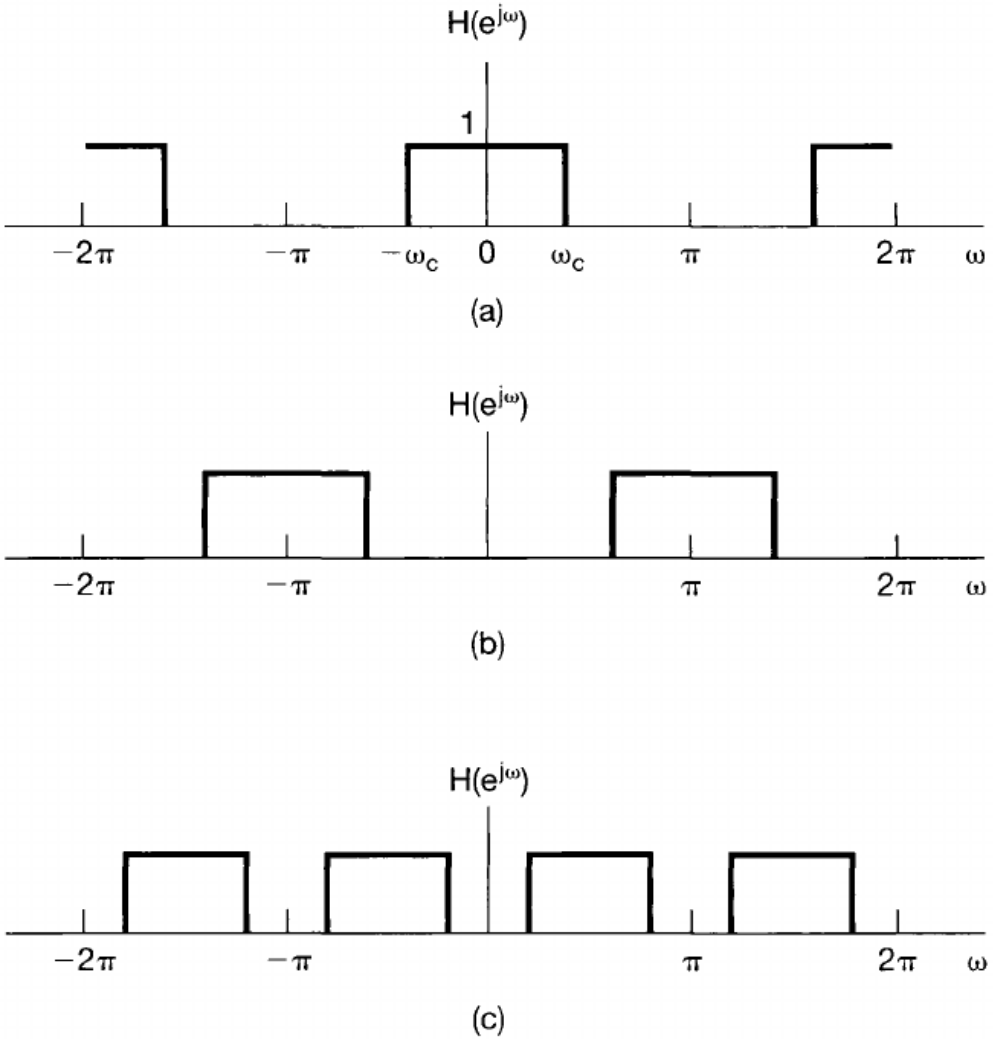


Image: Oppenheim 3.28

Example: filters

Determine the impulse response of an ideal DT low-pass filter,

$$H(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| < \omega_c \\ 0, & \omega_c \leq |\omega| < \pi \end{cases}$$

Example: filters

For an ideal DT high-pass filter,

Properties of the DT Fourier transform

TABLE 5.1 PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Section	Property	Aperiodic Signal	Fourier Transform
		$x[n]$	$X(e^{j\omega})$ periodic with
		$y[n]$	$Y(e^{j\omega})$ period 2π
5.3.2	Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
5.3.3	Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$
5.3.3	Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$
5.3.4	Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
5.3.6	Time Reversal	$x[-n]$	$X(e^{-j\omega})$
5.3.7	Time Expansion	$x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$	$X(e^{jk\omega})$
5.4	Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
5.5	Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
5.3.5	Differencing in Time	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X(e^{j\omega})$
5.3.5	Accumulation	$\sum_{k=-\infty}^{n} x[k]$	$\frac{1}{1-e^{-j\omega}}X(e^{j\omega})$ $+\pi X(e^{j0})\sum_{k=-\infty}^{+\infty}\delta(\omega - 2\pi k)$
5.3.8	Differentiation in Frequency	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
5.3.4	Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \mathcal{R}e\{X(e^{j\omega})\} = \mathcal{R}e\{X(e^{-j\omega})\} \\ \mathcal{I}m\{X(e^{j\omega})\} = -\mathcal{I}m\{X(e^{-j\omega})\} \\ \lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert \\ \measuredangle X(e^{j\omega}) = -\measuredangle X(e^{-j\omega}) \end{cases}$
5.3.4	Symmetry for Real, Even Signals	$x[n]$ real an even	$X(e^{j\omega})$ real and even
5.3.4	Symmetry for Real, Odd Signals	$x[n]$ real and odd	$X(e^{j\omega})$ purely imaginary and odd
5.3.4	Even-odd Decomposition of Real Signals	$x_e[n] = \mathcal{E}v\{x[n]\}$ $[x[n]$ real$]$ $x_o[n] = \mathcal{O}d\{x[n]\}$ $[x[n]$ real$]$	$\mathcal{R}e\{X(e^{j\omega})\}$ $j\mathcal{I}m\{X(e^{j\omega})\}$
5.3.9	Parseval's Relation for Aperiodic Signals	$\sum_{n=-\infty}^{+\infty}\lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi}\lvert X(e^{j\omega})\rvert^2 d\omega$	

Image credit: Oppenheim Table 5.1

Properties of the DT Fourier transform

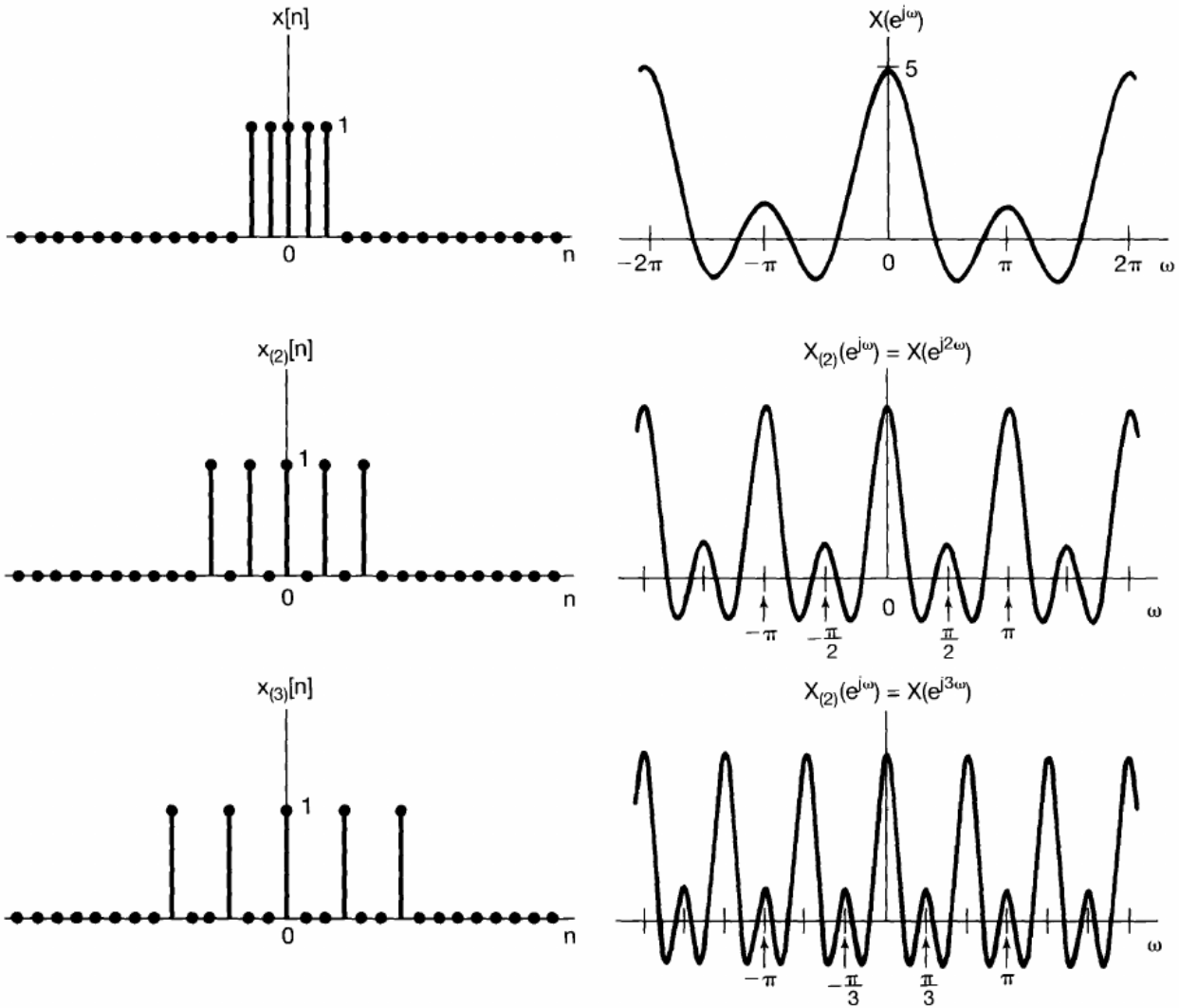
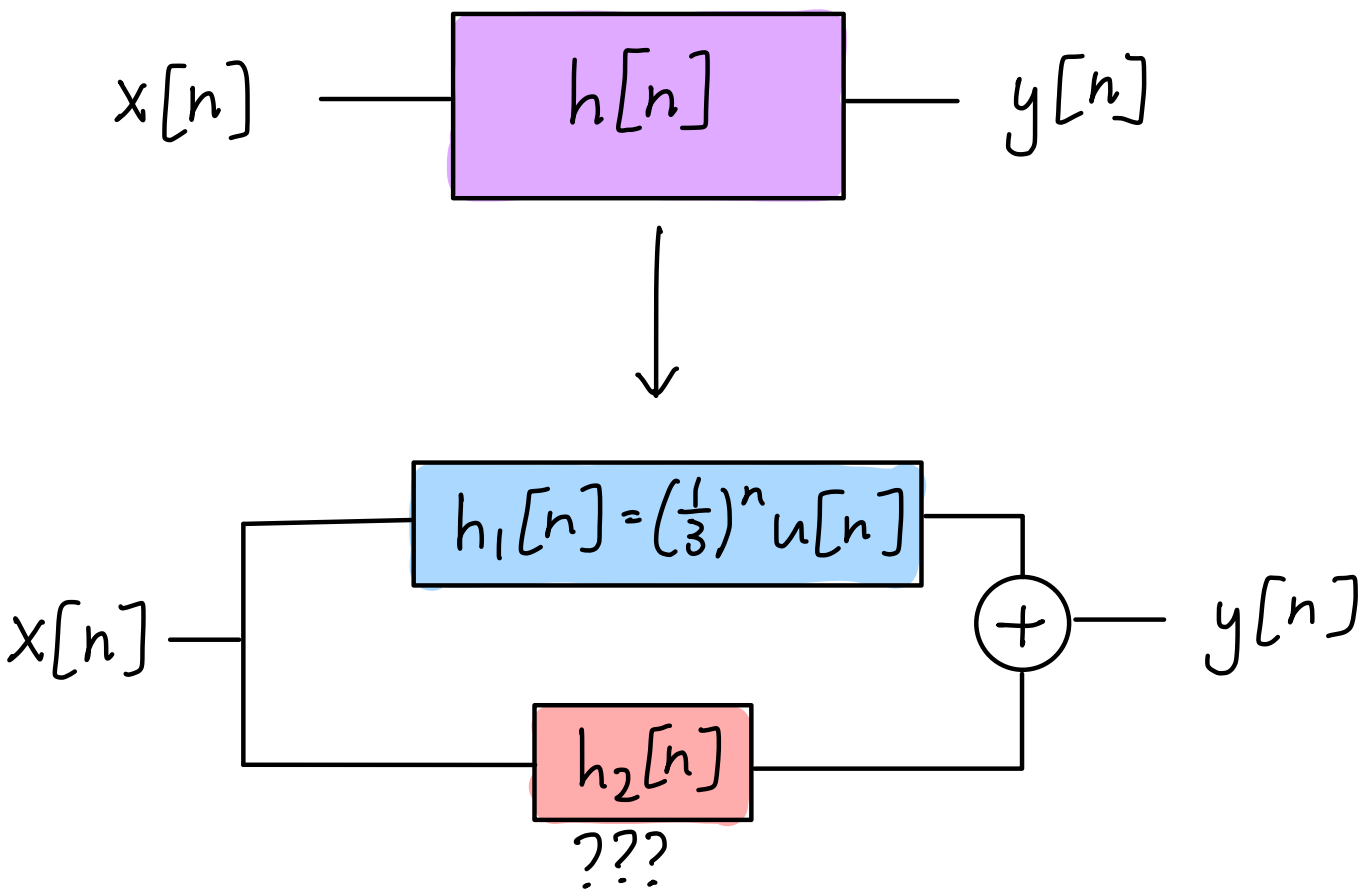


Image credit: Oppenheim Figure 5.14

Example

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$



Example

What is the DTFT of

$$x[n] = a^n u[n], \quad |a| < 1$$

Example

Properties of the DT Fourier transform

Time shift:

Differencing:

Example: difference equations

What is the frequency response of a system defined by a linear, constant-coefficient difference equation

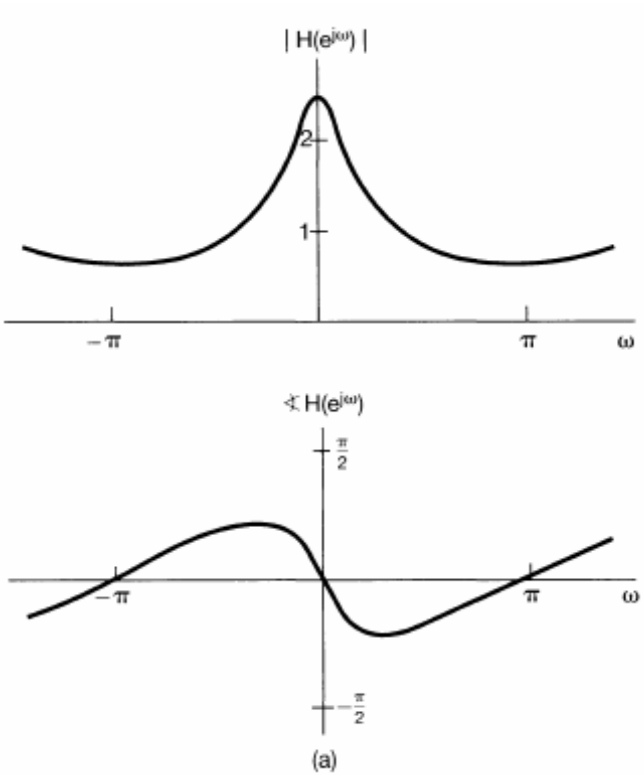
DT systems based on difference equations

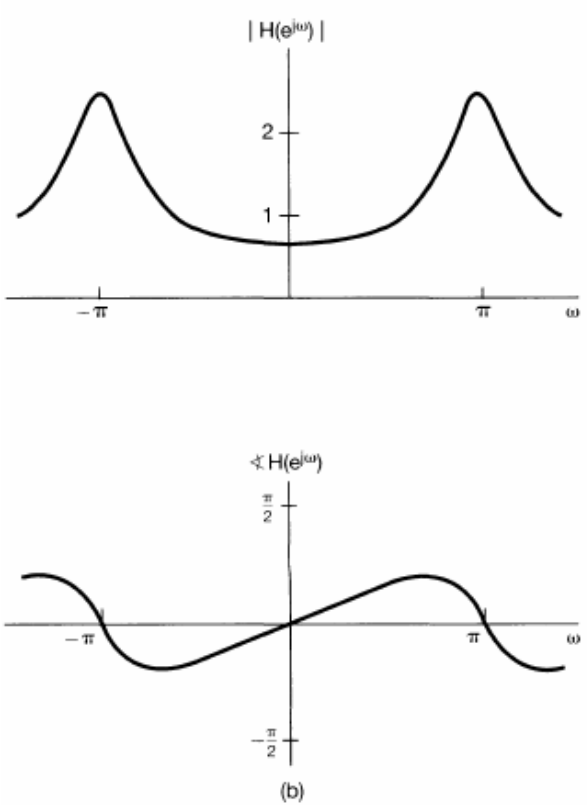
Exercise: compute frequency response of a system characterized by

$$y[n] - ay[n-1] = x[n]$$

Example

"First-order recursive DT filters"





$0 < a < 1 \ (a = 0.6)$

$-1 < a < 0 \ (a = -0.6)$

Image credit: Oppenheim Figure 3.34

DT systems based on difference equations

Exercise: compute frequency response of a system characterized by

$$y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$$

DT systems based on difference equations

Exercise: compute frequency response of a system characterized by

$$y[n] = \frac{1}{5}(x[n+2] + x[n+1] + x[n] + x[n-1] + x[n-2])$$

DT systems based on difference equations

Exercise: compute frequency response of a system characterized by

$$y[n] = \frac{1}{N+M+1} \sum_{k=-N}^{M} x[n-k]$$

DT systems based on difference equations

"Non-recursive DT filters"

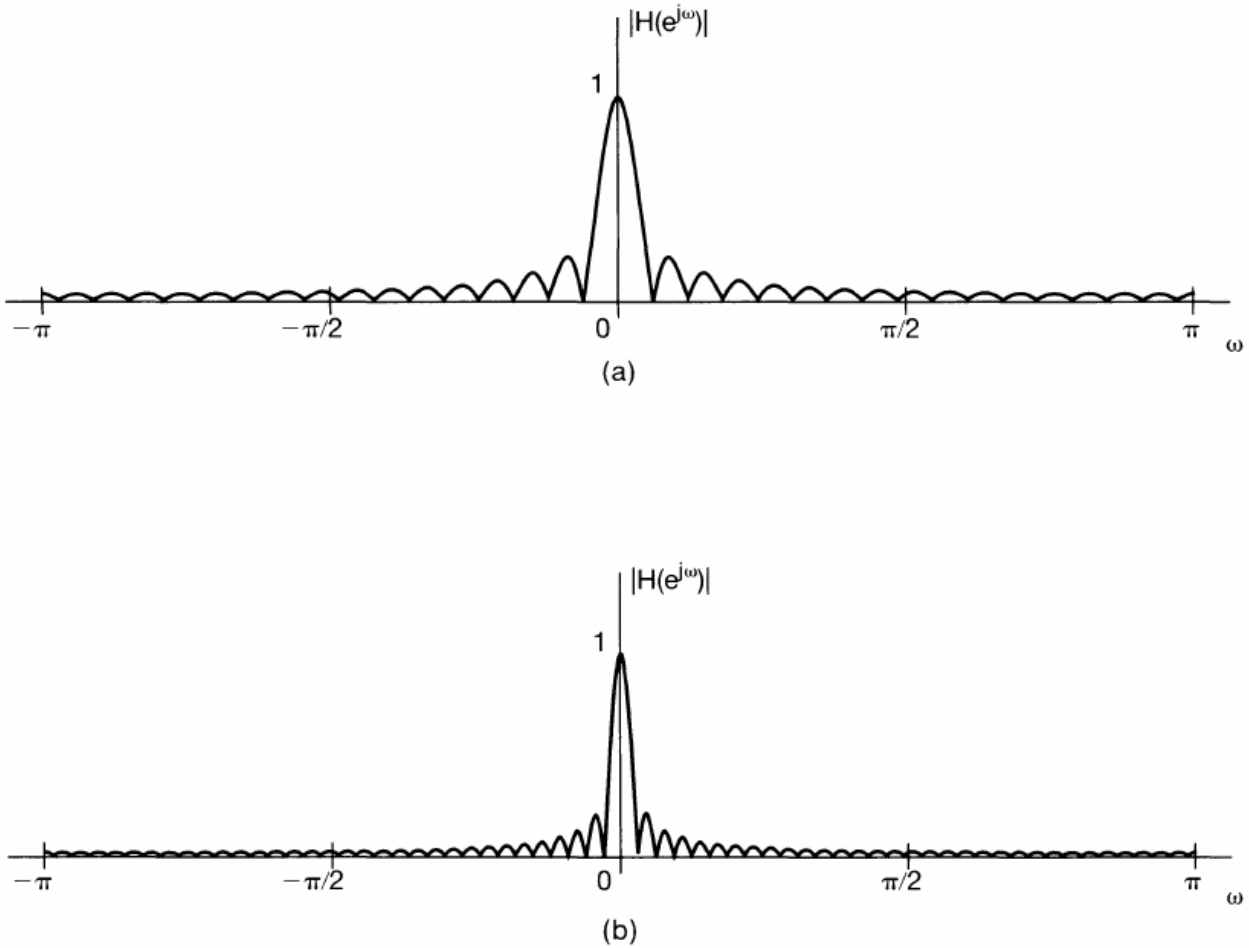


Image credit: Oppenheim Figure 3.36

For next time

Content:

- The sampling theorem
- Basics of interpolation
- The Nyquist rate and aliasing

Action items:

1. Enjoy not having to study for midterm

Recommended reading:

- From this class: Oppenheim 5.2-5.9
- Suggested problems: 5.4b, 5.6, 5.8, 5.19, 5.22bcdfgh, 5.25, 5.29, 5.31, 5.33-5.36
- For next class: Oppenheim 7.1-7.3